Measurement Science and Technology

PAPER

A versatile system for Hall effect measurements at high temperature

To cite this article: Murat Gunes *et al* 2017 *Meas. Sci. Technol.* **28** 105905

View the article online for updates and enhancements.

Related content

- An instrument for the high temperature measurement of the Seebeck coefficient and electrical resistivity
Murat Gunes, Mehmet Parlak and Macit Ozenbas
- Experimental setup for measurements of transport properties at high temperature and under controlled atmosphere
Céline Byl, David Bérardan and Nita Dragoe
- Low temperature Hall measurements in a cryostat with a built-in magnet
M Altwein, H Finkenrath and T Stockel

This content was downloaded from IP address 129.175.97.14 on 15/12/2017 at 23:35

IOP Publishing Measurement Science and Technology
Meas. Sci. Technol. **28** (2017) 105905 (8pp) https://doi.org/10.1088/1361-6501/aa7fe1

A versatile system for Hall effect measurements at high temperature

Murat Gunes, Hisanori Sato, Loreynne Pinsard-Gaudart, David Berardan and Nita Dragoe

Université Paris Sud, Université Paris Saclay, ICMMO, SP2M, UMR 8182 CNRS, 91405 Orsay, France

E-mail: nita.dragoe@u-psud.fr

Received 4 April 2017, revised 12 July 2017
Accepted for publication 14 July 2017
Published 14 September 2017



Abstract

We present a fast Hall effect measurement system that can be used at high temperature. The use of a homogeneous high field permanent magnet in a Halbach configuration allows fast measurements in various DC and AC current fields with step and continuous measurement modes. The results are presented of measurements on platinum film at room temperature and Ge and BiCuSeO between 300 K and 650 K.

Keywords: Hall effect, Halbach magnet, galvanomagnetic

S Supplementary material for this article is available online

(Some figures may appear in colour only in the online journal)

1. Introduction

Hall effect measurements are essential for assessing the type of charge carrier as well as its concentration and, coupled to the measurement of electrical conductivity, allow the determination of a carrier's mobility [1, 2]. Several systems have been described in the literature [2–26], for the largest part involving measurements at room or low temperature. Measurements at high temperature are less common [27–42], which is related to general difficulties in instrumentation at high temperature, including materials stability, thermal gradients, thermoelectric effects and thermal fluctuations during the measurement. Despite these difficulties, Hall effect measurements at high temperatures are valuable, as they can contribute to the understanding of electrical transport properties that are sensitive to temperature. Some approaches and caveats towards high temperature Hall effect measurements have been described in recent papers [27, 28]. Some of the previously mentioned difficulties can be avoided by using an AC current and a DC magnetic field ([28] and references therein) but the remaining problems require careful attention. For instance, Seebeck voltage, thermal fluctuations or changes to the magnetic field during the measurement, imprecise geometry of the electrical contacts, etc, can cause significant errors, especially in low mobility samples, due to the decrease in the signal to noise ratio at high temperature [1]. Besides, in usual systems, one measurement can take several tens of seconds, and the mitigation of temperature evolution effects requires the design of very stable temperature systems [33]. Thus, the ability to take measurements quickly would be an obvious advantage.

In this study we describe a new high-temperature Hall effect measurement system. This system uses a DC or AC field with either an AC or DC current source and is based on selecting specific orientations of the magnetic field relative to the sample. By using an AC field under AC excitation, our system can be used to measure the Hall effect continuously, providing rather rapid measurement. This possibility might be of great interest as it enables the monitoring of the evolution of the carrier concentration of a sample at high temperature, for example during a thermal treatment. The system we will describe here can also be used to measure magnetoresistance or (as it is incorrectly called) the planar Hall effect [22, 23, 43, 44].

2. Technical background

Under a magnetic field of arbitrary orientation, the potential field is expressed in (1) as [45]:

$$\vec{\mathbf{E}} = \rho_{\perp}\vec{\mathbf{j}} + \vec{\mathbf{n}}\left(\vec{\mathbf{j}}\cdot\vec{\mathbf{n}}\right)\left(\rho_{\parallel} - \rho_{\perp}\right) + \rho_{\mathrm{H}}\vec{\mathbf{n}}\times\vec{\mathbf{j}} \qquad (1)$$

 © 2017 IOP Publishing Ltd Printed in the UK

where $\vec{\mathbf{n}}$ is the magnetic field vector, $\vec{\mathbf{j}}$ is the current density and $\rho_{||}$ and $\rho_{\perp}$ are the resistivities along and perpendicular to the magnetic field. By adjusting the direction of **B** relative to the sample and $\vec{\mathbf{j}}$, one can access the Hall components together with the magnetoresistance. For an isotropic sample in an arbitrary position, figure 1, with ψ the angle between the normal of the sample and the magnetic field, and φ the angle between $\vec{\mathbf{j}}$ and the projection of the field on the sample, the signal will have the form [43]:

$$V_{\mathrm{H}} = \frac{\mu_{\mathrm{H}}\rho}{d} IB\cos\psi + \frac{C\mu_{\mathrm{H}}^2\rho}{2\,d} IB^2 \sin 2\varphi \sin^2\psi. \qquad (2)$$

Here, ρ is the material resistivity, d is the sample thickness, I is the applied current, C is a constant depending on the relaxation time, and μ_{H} represents the Hall mobility. We note, however, that the magnetoresistance is a quadratic function of the field [17, 44] and therefore any improper alignment of the sample can preclude the measurement of the Hall voltage, particularly for high magnetoresistance samples.

One of the problems in measuring the Hall effect is to do with the parasitic voltages (or so-called offsets), which can be more than several orders of magnitude larger than the Hall voltage, depending on the samples. This is particularly relevant for samples with low mobility. Several techniques have been proposed to cancel these offsets: field reversing [9, 28, 46], continuous spinning current [12, 13], dual AC sources [11, 14, 19, 24, 25], AC field [15, 38, 47], etc. Using an AC field is of particular interest due to the fact that offsets are in general of the DC type, so they can be filtered out with a lock-in amplifier.

One additional parameter to be taken into account is the type of current source; if a DC current source is used, an inversion of the current is required in order to cancel out Seebeck voltages that may be present in a sample if a temperature gradient exists [9]. Indeed, thermoelectric electromotive force (emf) caused by the temperature gradient across the specimen at high temperatures can result in a significant measurement error [10, 13, 30, 48]. For instance, the thermoelectric signal for a temperature gradient as low as 50 mK of a 'good' thermoelectric material can be in the few microvolts range and can be of the same order of magnitude as or larger than the Hall voltage. Therefore, the stability and the homogeneity of the temperature during each measurement for a DC current source system are essential.

Figure 2 is a diagram of the measurement system, including all modes of measurement. The violet dashed lines refer to the AC measurement mode and the brown dashed lines represent the DC measurement option. Since different kinds of measurements require different instruments, it is sometimes necessary to switch the connection from one measurement mode to another. For both modes, the same GPIB/Ethernet data acquisition setup and LabView software is used. The advantage of an AC source is that the possible Seebeck voltage is cancelled out without any need for current switching, and any small evolution of the temperature gradient during measurement will have little effect. We selected a system with an AC and DC current source (Keithley 6221), a lock-in amplifier (Stanford Research System 830 or Zurich Instruments MFLI), a nanovoltmeter (Keithley 2182), and either an AC or DC magnetic field, the orientation of which is controlled relative to the sample, corresponding to vector **n** in equation (1). A photograph of the whole system is presented in supplementary information figure S1 (stacks.iop.org/MST/28/105905/mmedia).

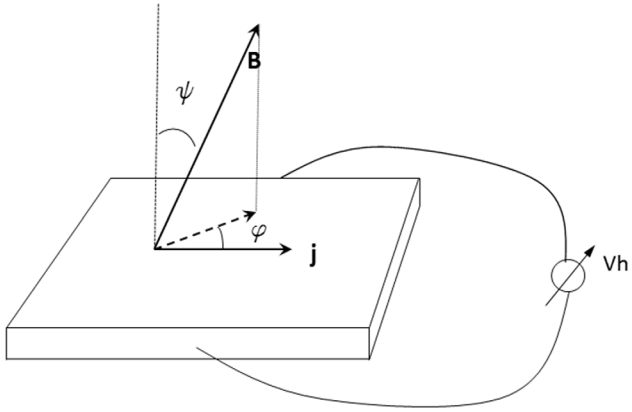


Figure 1. A sketch of angle and direction in relation to sample, current and magnetic field, referring to equation (2).

3. System

3.1. Magnetic field

The originality of the implementation of the setup described hereafter is the magnet. We selected a donut-shaped magnet with a hole of diameter 30 mm, in a Halbach array configuration, as shown in figure 3 [26]. In our case, it consists of eight $Nd_2Fe_{14}B$ dipoles oriented so as to obtain a homogeneous magnetic field induction of 1 T oriented in a radial direction in the center of the 'array' (figure 3). The homogeneity of the field in the center of the array is better than 3% in a 0.5 cm^3 volume, in our case. Larger volumes are also possible with this configuration. For instance, a Halbach with a magnet diameter size of 1 m was built for an MRI device for the international space station [49]. The Halbach magnet was chosen here for several reasons: field homogeneity and intensity, and in particular because it enables a short measurement time, as the orientation of the field can be changed quickly and in a controllable manner, simply by rotating the array with a motor.

The magnet was fixed on a motorized stage from Owis. The motor can be rotated up to about 2 Hz or fixed at a given position. A hole of diameter 28 mm inside the motor can be used for cooling the sample stage (see later).

3.2. Sample holder

The sample probe with all components is shown in figure 4. The sample (10) is mechanically fixed on a 1.5 mm thick aluminum nitride plate (AIN) (9) that is placed on a 16 mm diameter copper furnace (8). The electrical connection between the electronics and the sample is made with copper wire (4). An incoloy 50 W commercial cartridge heater (7) of 3.2 mm diameter is mechanically fixed inside the copper furnace, at the base. The sample probe is inserted into a Pyrex tube (6) that is fixed to a flange (5) which is attached to the holding

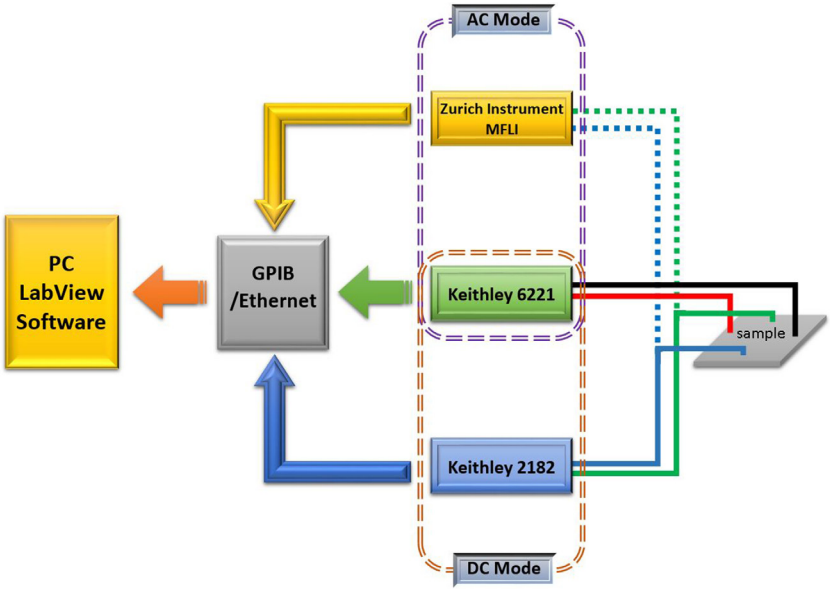


Figure 2. Block diagram of the Hall effect measurement system. The current source is common to both setups with either a nanovoltmeter for DC mode or a lock-in amplifier for AC mode.

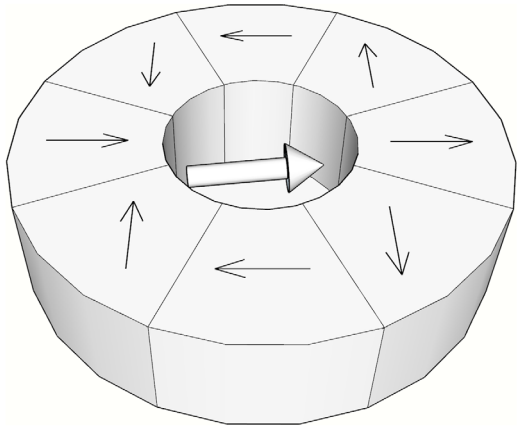


Figure 3. A sketch of the Halbach array used. The direction of the resultant magnetic field induction is indicated by the arrow in the center of the magnet.

platform (2). Finally, the holding platform is fixed to the sliding platform (1) using screws (2), which moves up and down for better positioning inside the magnet (11). The step motor (12) is placed under the magnet, and the air inlet (13) for cooling is attached to the hole passing through the step motor and the magnet. A K-type thermocouple is placed just behind the sample (see figure 5) on the AIN plate in order to measure and control the temperature during the measurement. We note that the thermal conductivity of the AlN plate is larger than 170 W K^{-1} m^{-1}, thus a good thermal homogeneity is ensured around the sample. However, in this setup the AlN holder is thermally low in its connection to the sample, so stabilization of the temperature is rather long (a matter of minutes). This is not important in this case since the duration of a measurement is very short (a matter of seconds) and the measurements are made during a temperature sweep (see later). A better sample holder might be built by using a complete furnace made from a good thermal conducting ceramic such as boron nitride, shapal, or aluminium nitride.

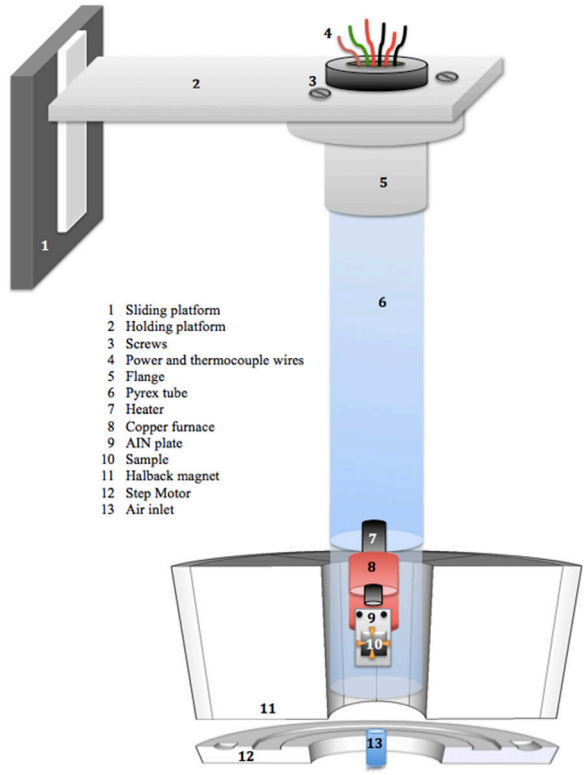


Figure 4. The sketch of the sample probe with all parts numbered, and listed top left.

A detailed illustration of the sample probe with all components and contact types is shown in figure 5. Copper wires pass through alumina isolators inside the copper furnace and then to the AIN plate. They are mechanically fixed to brass screws. The contacts with the sample are made through triangular copper-beryllium alloy pins that are clamped with the help of screws.

In this setup, the maximum temperature of the measurements is limited to 773 K. Higher temperatures may be

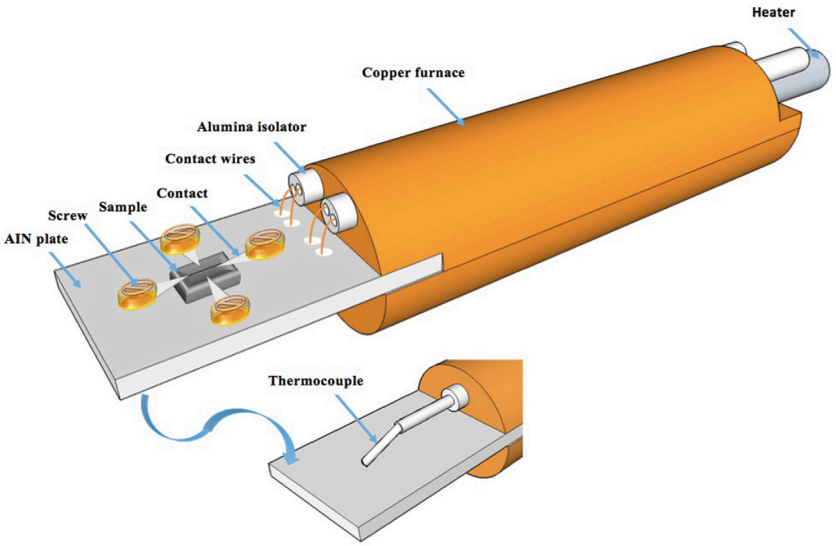


Figure 5. Illustration of contact configuration. Contacts are made on the surface of the sample via triangular Cu–Be pins. The AIN plate is firmly attached to the copper furnace. Contact wires are mechanically attached to the screws. A heater is inserted into the copper furnace and the thermocouple is placed behind the sample on the AlN plate.

possible, depending mainly on the diameter of the magnet. The sample holder is fixed on an inconel tube of 20 cm length and 6 cm diameter, through which electrical contacts are passed to the sample. A Pyrex tube of diameter 22–28 mm that passes through the magnet (but not through the motor) is fixed around the sample, in order to insert the sample in the magnetic field. The sample is then kept under vacuum ($1 \cdot 10^{-5}$ mbar) during the temperature-dependent measurements. Air-cooling is passed through the motor and inside the Halbach array in order to cool the magnet. This cooling system that maintains the magnet at room temperature is essential, since heating of the magnet would decrease the field and could eventually damage it, if the temperature were too high.

3.3. Connections

Four electrical contacts made with inconel springs in a cross configuration to the sample (similar to those described in a previous report [28]) are connected to the current source (Keithley 6221) and a lock-in amplifier (Stanford Research System 830 or Zurich Instrument MFLI), or to a Keithley 2182 nanovoltmeter in the case of DC measurements. When measuring in an AC field, an optical detection of the position of the magnet is fed as the reference for the lock-in amplifier. In a DC field the position is read through the internal Hall effect sensors of the motor. The temperature of the sample is controlled with a Eurotherm 2408 controller. All equipment is controlled via GPIB or Ethernet interfaces by a LabView program developed in our laboratory.

3.4. Measurements

For the Hall effect, we implemented six types of measurements: either fixed (noted DC) magnetic field or alternate field (called hereafter AC field), and for each field one can use DC or AC current. For the last, DC and AC measurements are conducted using step or scanning modes. In the following, we will describe the measurements as: DC field/DC current as DC/DC; DC field/AC current as DC/AC; AC field/AC current as AC/AC and AC field/DC current as AC/DC. The measurement does not necessarily require the magnetic field to be perpendicular to $\vec{\mathbf{j}}$. As mentioned previously (see (1) and (2)), the Hall voltage is then directly proportional to cos (ψ) with ψ the angle between the normal to the sample and the N-pole of the magnetic field.

3.4.1. DC field measurements. In this configuration, the magnet is rotated to a given angle and a measurement is taken. As the Hall voltage V_H has cosine dependence on the angle, the slope of the V_H versus cos (ψ) directly gives the Hall constant. Because of the large number of points that can be recorded (as we use typically 8–20 points, a measurement takes only a few seconds) the V_H signal can easily be extracted. An example is given in figures 6(a) and (b). Note that due to various offsets (and wire inductance) there is an offset in the Hall voltage, but our measurements show that signals two orders of magnitude smaller than the offset are very well observed. In this example, the sample was a platinum foil of 0.05 mm thickness and the measurement was made with a current of 0.07 A rms at a frequency of 47 Hz. The signal, of the order of 5×10^{-8} V, can be easily extracted from the offset, which is larger than 7.7×10^{-6} V. Continuous measurements for a period of over 48 h taken from the same sample consistently give the same results (see supplementary information figure S2).

The measurement of the Hall signal can be made either in DC mode, using the Keithley 6221 AC/DC source and with the Keithley 2182 nanovoltmeter in 'delta' mode [50], or in AC current mode and measuring with a lock-in amplifier. We did not notice any significant difference between the two measurement methods. The average Hall coefficient value for platinum as measured by this system was $-0.21 \cdot 10^{-12}$ Ωcm/Gauss with a standard deviation of $\pm 0.01 \cdot 10^{-12}$ Ωcm/Gauss at room temperature, which is similar to reported values ($-0.21 \cdot 10^{-12}$ Ωcm/Gauss [51], $-0.24 \cdot 10^{-12}$ Ωcm/Gauss

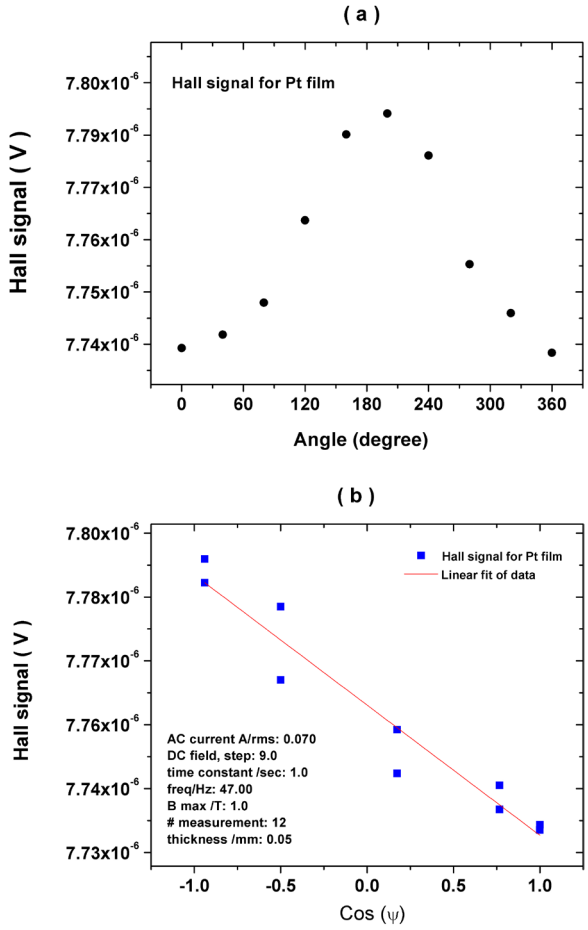


Figure 6. (a) Hall signal for a given angle of the magnet for a metallic Pt film; and (b) the slope of the obtained data versus cos (ψ).

[52], and $-0.23 \cdot 10^{-12}$ Ωcm/Gauss [53]). Extensive measurements for metals such as Cu, Ta and Mo gave Hall coefficients consistent with reference values (not shown here).

Alternatively, the magnet can be rotated continuously at a speed of up to about 0.2 to 0.5 Hz while the measurements are performed continuously with the nanovoltmeter or with a lock-in amplifier. The continuous method is faster but its applicability depends on the time constant used for the measurement, which should be small in comparison with the rotation speed. Examples of two 'dynamic' measurements for a Ge (1 1 1) sample in DC/AC scanning mode are shown in figure 7; the magnet is turned with 0.5 Hz and 0.2 Hz and measurements are taken for a full rotation of the magnet in 2 s and 5 s.

For our system, the fastest measurement is about 1 s to 2 s; this is related mainly to the maximum rotation speed of our magnet.

We note that a shift of the signal might be observed, which is related to the measurement speed relative to the rotation speed. An example of such a shift is shown in supplementary information, in figure S3. This shift can be minimized by decreasing the integration time (in the case of figure 7 the lock-in time constant was much shorter than the rotation time, hence no shift of the signal can be observed). On the basis of this evidence for 'simple cases' such as semiconductors,

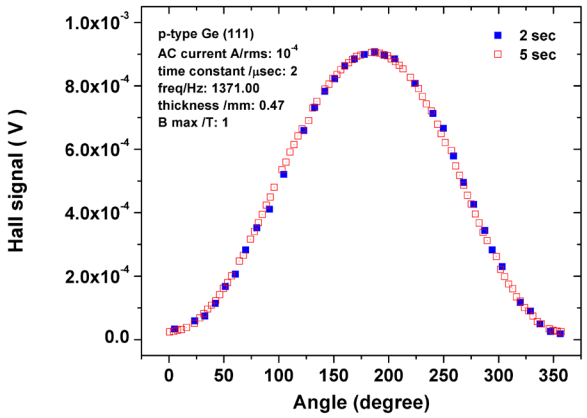


Figure 7. Data obtained (V_H signal) during the measurements at 2 s and 5 s for a p-doped Ge (1 1 1) sample, in DC/AC mode.

one measurement can last only one second, which enables the *in situ* observation of rapid evolutions of the Hall constant of the materials. An example of such measurement for a single crystal of p-type Ge (1 1 1) is shown in the following (see section 'High temperature measurement')

3.4.2. AC field measurements. Interest in Hall AC field measurements has been underlined in several references, and recently a commercial instrument has been developed using this method [47]. The system we describe in this article has the advantage of being simple, affordable, and with an AC field frequency that depends mainly on the speed of the motor. In the case of an AC field, there are two distinct measurement methods.

3.4.2.1. AC field/AC current. A Hall system using an AC current source and an AC magnetic field has been proposed before [11, 54]. The idea is to submit the sample to a current of frequency ω_1 and a magnetic field of frequency ω_2 while measuring the Hall effect at $\omega_1 - \omega_2$ or $\omega_1 + \omega_2$ frequency. Here, the magnet is turning at a constant, selectable speed, and the rotation frequency is detected with an optical sensor. For AC current and an AC magnetic field,

$$I = I_0 \cos(\omega_1 t + \varphi); \quad H = H_0 \cos(\omega_2 t). \tag{3}$$

Here, I, I_0, ω_1, t, and φ are current, current amplitude, current frequency, time, and a phase shift relative to the field, and H, H_0 and ω_2 are the field, its amplitude, and field frequency, respectively. The Hall voltage is:

$$V_H = \frac{R}{2d} I_0 H_0 \{\cos[(\omega_1 - \omega_2)t + \varphi] + \cos[(\omega_1 + \omega_2)t + \varphi]\}. \tag{4}$$

In this case, the Hall coefficient can be obtained by measuring the voltage at $\omega_1 - \omega_2$ or at $\omega_1 + \omega_2$ frequency by using the power of lock-in amplification. The Hall constant is then given by [54]:

$$R_h = 2\, dV_H / IH \tag{5}$$

where d is the sample thicknes, I is the current and V_h is the measured voltage.

While higher frequency results in lower noise, the relative small frequency shift requires the use of high roll-over

Table 1. All measurement modes of implemented instruments with specifications and obtained results for a single crystal of p-type Ge (1 1 1).

Measurement mode	Current (A, peak)	Magnetic field (Hz)	Current frequency (Hz)	Scan speed (Hz)	Steps	Measurement duration (s)	Hall coefficient $\times 10^{-5}$ (Ωcm/Gauss)
DC Field, DC current step mode	$1 \cdot 10^{-3}$	DC	DC	—	19	20	1.928 ± 0.057
					9	10	1.913 ± 0.082
DC Field, DC current scan mode	$1 \cdot 10^{-3}$	DC	DC	0.025	—	40	1.899 ± 0.011
				0.050		20	1.870 ± 0.028
				0.100		10	1.851 ± 0.050
				0.200		5	1.887 ± 0.028
DC Field, AC current step mode	$1 \cdot 10^{-3}$	DC	17	—	9	30	1.881 ± 0.050
					11	60	1.901 ± 0.031
DC Field, AC current scan mode	$1 \cdot 10^{-3}$	DC	17	0.025	—	40	1.882 ± 0.010
				0.050		20	1.869 ± 0.024
				0.100		10	1.844 ± 0.044
				0.200		5	1.885 ± 0.069
AC Field, DC current mode	$1 \cdot 10^{-3}$	1.32	DC	—	—	40	1.820 ± 0.028
AC Field, AC measurement mode	$1 \cdot 10^{-3}$	1.32	16.5	—	—	30	1.911 ± 0.003

filtering with a lock-in amplifier, which requires longer acquisition times. We use typically a 1–2 Hz rotation speed and 3–18 Hz AC source. An example of measurement for the Ge sample with AC/AC mode is provided in table 1.

3.4.2.2. AC field /DC current. A different approach is to use a DC current source and an AC field. A similar system was recently described by Chen *et al* [15, 55] and their remarkable results underline the importance of an AC Hall effect measurement system. The AC signal is dependent on the magnetic properties of the sample and also on the sample holder, the latter being independent of the current used. Measurements taken both with and without current permit the Hall coefficient to be obtained, even for very low mobility materials [47].

Examples of the results obtained by these methods, all implemented in our instruments, are given in table 1. As can be seen from the table, all the results obtained using different methods under different conditions are consistent with each other. In this table we show results obtained for a p-type Ge single crystal with typical values for each measurement. For AC field measurements the magnetic field frequency was 1.32 Hz. As seen from the table, values calculated from different measurement modes are similar.

4. High temperature measurement

For measurements at higher temperature we used a Eurotherm 2408 controller to control the heater and tune the temperature of the sample holder. Generally, high-temperature measurements can be conducted either step-by-step or by continually ramping the temperature. Although the step-by-step mode might provide more accurate results due to temperature stability, it is a time-consuming type of measurement. As an example, a measurement was carried out in DC/DC mode with currents ranging from 10^{-6} A at room temperature to 10^{-4} A at high temperature. The n-type doped ($1 \cdot 10^{15}$ cm^{-3}) silicon single crystal sample with a thickness of 0.4 mm was mechanically attached to the sample probe using inconel 750 wires [28]. The result of this measurement with temperature steps is shown in the supplementary information (figure S4).

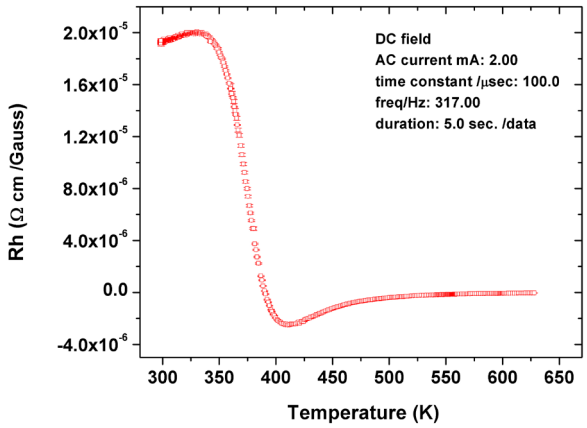


Figure 8. The p-type Hall coefficient as a function of temperature for a single crystal Ge (1 1 1) sample. The measurement duration is 5 s and applied current is 2 mA.

As can be seen from figure 7, our system can measure the Hall coefficient in a few seconds, therefore measurement during temperature evolution is possible. As an example, a high-temperature Hall coefficient measurement was conducted on a p-type Ge (1 1 1) sample (see figure 8). The measurement was conducted between 300 K and 625 K at a heating rate of about 300 K h^{-1} while continuously measuring the Hall coefficient. During this measurement the recording of each data point required only 5 s. The total measurement duration was about 75 min. Note that in a classical measurement system one temperature point takes at least several minutes (typically 10 min for one temperature).

In order to test samples of interest for our laboratory, thermoelectric materials, we show the results of a sample of Sr-doped

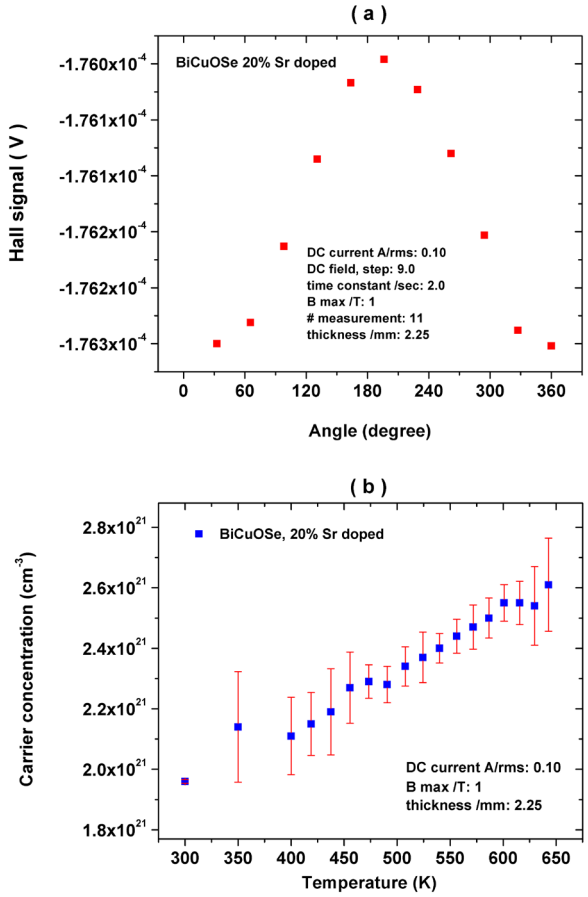


Figure 9. Hall signal for a given angle of the magnet for 20% Sr-doped BiCuSeO in DC/DC mode (a), and the carrier concentration value as a function of temperature (b).

BiCuSeO (figure 9), which is a promising material for high temperature thermoelectric energy conversion [56]. A bar of 2.25 mm thickness was measured with a current of 0.1 A. Room temperature carrier concentrations for samples of this composition were measured in a commercial system (PPMS), in a field range of −9 T to 9 T as described previously [57], and were about $2 \cdot 10^{21}$ cm^{-3}. As the mobility of this material is relatively low, $< 2\,\mathrm{cm}^2\,\mathrm{V}^{-1}\,\mathrm{s}^{-1}$, offsets related to contact positions are particularly difficult to overcome. For this particular case the offsets (10^{-5} V) were typically two orders of magnitude larger than the measured signal (10^{-7} V). The results obtained at high temperature are shown in figure 9. Every point in this figure is the mean value of at least 10 measurements at a given temperature.

The measurement of other galvanomagnetic properties [1, 45], for instance the planar Hall effect [43], is also possible with this setup. According to equation (1), selecting a particular angle between $\vec{\mathbf{n}}$ and $\vec{\mathbf{j}}$ can also permit measurements of the magnetoresistance. In our setup, the sample is placed in the plane of the 'donut'. Then, by selecting the proper direction of rotation of the magnet, either the Hall coefficient or the magnetoresistance can be determined.

5. Conclusions

We have presented a simple, versatile and affordable system for DC and AC field Hall effect measurements at high temperature that can also be used for other galvanomagnetic properties' measurements. Our results indicate that both metals and semiconductors can be easily studied by our system. For low temperature measurements the insertion of a cryostat inside a Halbach magnet would also possible, thus expanding the applicability of the setup.

References

[1] Lindberg O 1952 Hall effect *Proc. IRE* **40** 1414–9
[2] van der Pauw L J 1958 A method of measuring the resistivity and Hall coefficient of discs of arbitrary shape *Philips Res. Rep.* **13** 1–9
[3] Shewchun J 1971 A computer controlled automatic system for measuring the conductivity and Hall effect in semiconducting samples *Rev. Sci. Instrum.* **42** 1797
[4] Shapira Y and Foner S 1973 EuO. I. Resistivity and Hall effect in fields up to 150 kOe *Phys. Rev.* B **8** 2299–315
[5] Petritz R L 1958 Theory of an experiment for measuring the mobility and density of carriers in the space-charge region of a semiconductor surface *Phys. Rev.* **110** 1254–62
[6] Lomas R A, Hampshire M J and Tomlinson R D 1972 A sensitive method of Hall measurement *J. Phys. E: Sci. Instrum.* **5** 819
[7] Green R 1879 Hall effect measurements in materials characterization *White Paper* pp 1–12
[8] Guthrie G L 1965 Sensitive ac Hall effect circuit *Rev. Sci. Instrum.* **36** 1177–9
[9] Pell E M and Sproull R L 1952 Sensitive recording alternating-current Hall effect apparatus *Rev. Sci. Instrum.* **23** 548–52
[10] Olson E E and Wertz J E 1970 A high impedance AC Hall effect apparatus *Rev. Sci. Instrum.* **41** 419–21
[11] Kaneda T, Kobayashi S and Shimoda K 1973 A double AC method for sensitive measurement of the Hall effect *Japan. J. Appl. Phys.* **12** 1335–7
[12] Steiner R, Maier C, Häberli A, Steiner F and Baltes H 1998 Offset reduction in Hall devices by continuous spinning current method *Sens. Actuators* A **66** 167–72
[13] Munter P J A 1989 A low-offset spinning-current Hall plate *Sens. Actuators* A **22** 743–6
[14] Abbas D C and Phelps D J 1981 Double AC Hall effect system with simple, digital oscillator with application to semiconductor measurements *J. Phys. E: Sci. Instrum.* **14** 1078–80
[15] Chen Y, Yi H T and Podzorov V 2016 High-resolution ac measurements of the Hall effect in organic field-effect transistors *Phys. Rev. Appl.* **5** 1–9
[16] Fujime S, Murakami M, Hirahara E, Satoshi F and Miyuki Murakami E H 1961 Hall effect on some of the magnetic compounds *J. Phys. Soc. Japan* **16** 183–6
[17] Berlincourt T G 1959 Hall effect, resistivity, and magnetresistivity of Th, U, Zr, Ti, and Nb *Phys. Rev.* **114** 969
[18] Cevdet C *et al* 2004 An experimental set-up for *in situ* Hall measurements under high-energy electron irradiation for wide-bandgap materials *Meas. Sci. Technol.* **15** 297
[19] Chu P, Niki S, Roach J W and Wieder H H 1987 Simple, inexpensive double AC Hall measurement system for routine semiconductor characterization *Rev. Sci. Instrum.* **58** 1764
[20] Clayhold J A, Kerns B M, Schroer M D, Rench D W, Logvenov G, Bollinger A T and Bozovic I 2008 Combinatorial measurements of Hall effect and resistivity in oxide films *Rev. Sci. Instrum.* **79** 033908

[21] Lee Y S, Winkler M T, Siah S C, Brandt R and Buonassisi T 2011 Hall mobility of cuprous oxide thin films deposited by reactive direct-current magnetron sputtering *Appl. Phys. Lett.* **98** 192115

[22] Naftalis N, Haham N, Hoffman J, Marshall M S J, Ahn C H and Klein L 2012 Angular dependence of the Hall effect of La0.8Sr0.2MnO3 films *Phys. Rev.* B **86** 184402

[23] Ogrin F, Lee S and Ogrin Y 2000 Investigation of perpendicular anisotropy of a thin film using the planar Hall effect *J. Magn. Magn. Mater.* **219** 331–9

[24] McLevige W V, Chatterjeet P K and Streetman B G 1977 Versatile double AC Hall effect system for profiling impurities in semiconductors *J. Phys. E: Sci. Instrum.* **10** 335–7

[25] Kasai A, Abdulla A, Watanabe T and Takenaga M 1994 Highly sensitive precise double AC Hall effect apparatus for wide resistance range *Japan. J. Appl. Phys.* **33** 4137–45

[26] Raich H and Blümler P 2004 Design and construction of a dipolar Halbach array with a homogeneous field from identical bar magnets: NMR Mandhalas *Concepts Magn. Reson.* B **23** 16–25

[27] Adnane L, Gokirmak A and Silva H 2016 High temperature Hall measurement setup for thin film characterization *Rev. Sci. Instrum.* **87** 75117

[28] Borup K A, Toberer E S, Zoltan L D, Nakatsukasa G, Errico M, Fleurial J P, Iversen B B and Snyder G J 2012 Measurement of the electrical resistivity and Hall coefficient at high temperatures *Rev. Sci. Instrum.* **83** 0123902–7

[29] Ostrick B, Fleischer M and Meixner H 1997 High-temperature Hall measurements on $BaSnO_3$ ceramics *J. Am. Ceram. Soc.* **80** 2153–6

[30] Wood C, Lockwood A, Chmielewski A, Parker J and Zoltan A 1984 High temperature Hall effect apparatus *Rev. Sci. Instrum.* **55** 110–3

[31] Dauphinee T M and Mooser E 1955 Apparatus for measuring resistivity and Hall coefficient of semiconductors *Rev. Sci. Instrum.* **26** 660–4

[32] Rowe D M 1975 Electrical properties of hot-pressed germanium–silicon–boron alloys *J. Phys. D: Appl. Phys.* **8** 1092–103

[33] Lupu N Z, Tallan N M and Tannhauser D S 1967 Apparatus for measuring the Hall effect of low-mobility samples at high temperatures *Rev. Sci. Instrum.* **38** 1658–61

[34] Russell B R and Wahlig C 1950 A new method for the measurement of Hall coefficients *Rev. Sci. Instrum.* **21** 1028–9

[35] Bosman A J, Van Daal H J and Knuvers G F 1965 Hall effect between 300 K and 1100 K in NiO *Phys. Lett.* **19** 6–7

[36] Austin I G, Springthorpe A J, Smith B A and Turner C E 1967 Electronic transport phenomena in single-crystal NiO and CoO *Proc. Phys. Soc.* **90** 157–74

[37] Rowe D M and Bunce R W 1971 Apparatus for measuring resistivity and Hall coefficient of heavily doped semiconductors at high temperatures *J. Phys. E: Sci. Instrum.* **4** 902–4

[38] Perron J C 1970 Mesure du coefficient de Hall des metaux ou semiconducteurs liquides *Rev. Phys. Appl.* **5** 611–6

[39] Frank V 1959 Hall coefficient of technically pure metals from 80 K to 800 K *Appl. Sci. Res.* B **7** 41–51

[40] Hitchcock A H and Stringer J 1971 Temperature dependence of the Hall effect in α-phase copper-base alloys at high temperatures *J. Phys. D: Appl. Phys.* **4** 810–7

[41] McKinzie H L and O'Keeffe M 1967 High temperature Hall effect in cuprous oxide *Phys. Lett.* **24** 137–9

[42] Wang Y, Rogado N S, Cava R J and Ong N P 2003 Anomalous high-temperature Hall effect on the triangular lattice in Na_xCoO_2 **2** 4

[43] Schott C, Besse P and Popovic R S 2000 Planar Hall effect in the vertical Hall sensor *Sensors Actuators* A **85** 111–5

[44] Drabble J R and Wolfe R 1957 Geometrical effects in transverse magnetoresistance measurements *J. Electron. Control* **3** 259–66

[45] Jan J-P 1957 Galvanomagnetic and thermomagnetic effects *Solid State Phys.* **5** 1–96

[46] Lavine J M 1958 Alternate current apparatus for measuring the ordinary Hall coefficient of ferromagnetic metals and semiconductors *Rev. Sci. Instrum.* **29** 970–6

[47] Lindemuth J R An introduction to AC field Hall effect measurements *Lake Shore Cryotronics Report*

[48] Friend R H and Bett N 1980 Design of an alternating current source for resistivity and Hall effect measurements *J. Phys. E: Sci. Instrum.* **13** 294–7

[49] Sarty G E and André O 2012 Magnetic resonance imaging of astronauts on the international space station and into the solar system *Can. Aeronaut. Space J.* **58** 60–8

[50] Keithley 2012 6221 DC current source AC and DC current source *Datasheet* pp 1–5

[51] Greig D and Livesey D 1972 The Hall coefficient of dilute palladium and platinum alloys *J. Phys. F: Met. Phys.* **2** 699–708

[52] Gray D E 1957 *American Institute of Physics Handbook* (New York: McGraw-Hill)

[53] Hurd C M 1972 *The Hall Effect in Metals and Alloys* (Boston, MA: Springer)

[54] Bogomolov V N and Zhure V P 1962 Some modern methods of measuring the Hall effect *Sov. Powder Metall. Met. Ceram.* **1** 471–6

[55] Lee B, Chen Y, Fu D, Yi H T, Czelen K, Najafov H and Podzorov V 2013 Trap healing and ultralow-noise Hall effect at the surface of organic semiconductors *Nat. Mater.* **12** 1125–9

[56] Zhao L D, Berardan D, Pei Y L, Byl C, Pinsard-Gaudart L and Dragoe N 2010 $Bi_{1-x}Sr_xCuSeO$ oxyselenides as promising thermoelectric materials *Appl. Phys. Lett.* **97** 92118

[57] Barreteau C, Beardan D, Amzallag E, Zhao L and Dragoe N 2012 Structural and Electronic transport properties in Sr-doped BiCuSeO *Chem. Mater.* **24** 3168–78